Shruti Jagdale

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WORK EXPERIENCE

Software Engineer, ML Systems | Zeblok Computational | Stony Brook, NY — May 2025 - Jan 2026

- Owned model inference in production: served models with vLLM and GPU computing, profiled throughput
- Built failure detection and recovery, handled late or missing data, and kept a streaming pipeline reliable
- Built observability and dashboards with OpenTelemetry and Grafana to show run progress, slowdowns, and failures
- Did on-call support, troubleshooting guides, and capacity planning; cut API latency 25% with FastAPI
- Used AI and agentic coding tools to explore code and build fast; deployed with Docker, Kubernetes, and CI/CD on AWS

Graduate Research Assistant | Stony Brook University | Stony Brook, NY — Jan 2026 - May 2026

- Built a full-stack app with FastAPI, React, and PostgreSQL over 8.7M+ EHR events, handling missing data
- Built Transformer and RETAIN models in PyTorch to predict risk from longitudinal records

ML Engineer, Model Training Systems | L3Cube Research Lab | Pune, India — Jun 2023 - Jun 2024

- Ran GPU fine-tuning with PyTorch and visualized training runs, handling missing data in a 52K-pair dataset
- Published 3 papers in IEEE, ACL, and Springer

Data Scientist | FeyNN Labs | Assam, India — Nov 2022 - Jul 2023

- Built Python ETL pipelines for 10K+ IoT sensor records, handling missing data, and built models lifting accuracy 59%

EDUCATION

Stony Brook University | Stony Brook, New York — Aug 2024 - May 2026

Master of Science in Data Science, GPA: 3.78/4.00

Pune Institute of Computer Technology |Pune, India — Aug 2020 - Jun 2024

Bachelor of Engineering, GPA: 3.62/4.00

PROJECTS

Distributed Transaction Store | *Go, gRPC, Distributed Systems* — Sep 2026

- Built a sharded, fault-tolerant store in Go with Multi-Paxos consensus, leader election, and log recovery
- Wrote chaos tests in CI that kill nodes to verify failure recovery and consistency; shipped a gRPC SDK and AWS cluster

Atlas: Full-Stack Multi-Agent Travel Assistant | *LangGraph, Weather Data, Docker* — Jun - Sep 2026

- Built a LangGraph supervisor with parallel sub-agents using weather data, plus failure recovery and audit logs
- Split services into independently scalable Docker containers behind nginx, built with AI and agentic coding tools

Order Book Prediction | *Real-Time Data, Keras* — Feb - May 2025

- Processed real-time Binance order book data, filtering 40,493 raw updates to 36,816 valid snapshots
- Modeled buy pressure with a BiGRU in Keras using walk-forward TimeSeriesSplit

Terrorism Risk Forecasting | *Geospatial, R Shiny* — Feb - May 2025

- Analyzed geospatial data from the Global Terrorism Database (1970-2017) with an R Shiny map and dashboards
- Built ETS forecasts, logistic regression, and random forest models with SMOTE for class imbalance

Seller Support Copilot | *Reliability, Observability, AWS* — Sep 2026

- Built a FastAPI and Next.js app with failure detection and recovery, traced by OpenTelemetry, CloudWatch, and X-Ray
- Tested with pytest and Playwright; deployed on ECS Fargate with GitHub Actions CI/CD

MCP Server Trust Exploitation Module - DeepTeam (Confident AI) | *Debugging, Open Source* — Sep 2026

- Fixed a judge-evaluation bug spanning several modules of an open-source framework: zero regressions in 234 tests
- Built a module with 3 MCP security sub-types and opened an upstream pull request

Computer-Use Automation Agent | *Agent Tracing, Observability* — Aug 2026

- Built an agent logging every action as JSONL traces with an error taxonomy to show failures

PUBLICATIONS

Code-Mixed Embeddings for Hate Speech Detection | *Springer Nature, ICCCN 2024* — Jun 2025

Adapting LLMs to Low-Resource Settings | *ACL, CHiPSAL 2025* — Jan 2025

Code-Mixing in Named Entity Recognition | *IEEE, ICETITE 2024* — Feb 2024

TECHNICAL SKILLS

Languages/Tools: Python, PyTorch, Go, TypeScript, SQL, FastAPI, React, Docker, Kubernetes, AWS, vLLM, GPU Computing, R Shiny

ML Systems: Reliability, Failure Recovery, Real-Time Pipelines, Observability, Dashboards, Geospatial Data, Weather Data, Agentic Tools